Exe Flood Tool Report

This project integrates comprehensive EDA and preprocessing, robust modelling and prediction pipelines, and spatial visualisation to assess UK flood risk. Additional analytical interfaces extend the tool's capability, ensuring a complete, reliable, and assessment-ready solution.

EDA & Preprocessing

We conducted data preprocessing across datasets: For postcodes_labelled.csv, we converted coordinates, added postcode details, and imputed missing data via postcode and KNN methods. Stations.csv was split into rainfall and water level stations, with water level stations cleaned for metric order and missing values imputed by median. District_data.csv handled pet outliers via IQR and used a Random Forest to impute property age. For wet_day/typical_day.csv, we retrieved station coordinates, cleaned value columns, and produced preprocessed outputs.

Modeling & Prediction

Seven Class Flood Probability: A Random Forest classifier was employed to predict seven-class flood risk, using a unified preprocessing pipeline and KNN-based spatial matching. The model was evaluated against simpler baselines such as Logistic Regression and KNN classification, which underperformed on accuracy and non-linear patterns. Elevation, soil type and distance to watercourses emerged as the most informative features, while postcode-sector metadata and historic flooding contributed little due to sparsity.

Historic Flooding Classification: A Random Forest classifier was developed for historic-flooding prediction and evaluated against a Decision Tree baseline. Using postcode-level labelled data with unified preprocessing, stratified 5-fold cross-validation showed that the Random Forest achieved higher accuracy and substantially improved F1-scores, indicating a more reliable representation of historic flooding patterns.

Median House Price Predication: A combination of unit and sector data were selected as the features. A robust scaler was applied to deal with the heavily skewed numerical features, and OneHotEncoding was applied only on the soilType with the rest of categorical features dropped due to the high number of unique values. A modified XGBoost regression model was utilized for this prediction due to (1) its ability to use a validation set during training, (2) good performance in terms of prediction and training time, and (3) the well-balanced training vs validation fit. Overall, the most significant improvement was due to applying a log-transformer on the target variable, which resulted in a reduction in the RMSE, and MAE errors. This is expected because the median house price had significant outliers (up to ~8 million GBP).

Arbitrary location analysis: We utilized KNN models to predict flood risk and Local Authority for arbitrary coordinates. Through Grid Search Cross-Validation, we optimized classification (k=5) and regression (k=15, distance-weighted) parameters. Relying solely on OSGB36 coordinates, our spatial models significantly outperformed the baseline by leveraging spatial autocorrelation. Additionally, we implemented a meshgrid algorithm to visualize continuous flood risk across the UK.

Other Interfaces: estimate_total_value (calculates via median house price × household count), estimate_annual_human_flood_risk (computes using population, flood probability & impact coefficient), estimate_annual_economic_flood_risk (determines via total property value, flood probability & damage coefficient), *predict_high_risk_near_watercourses* (predicts flood risk levels by coordinates, filters high-risk areas near watercourses for specified postcodes)

Visualization

Figure 1 presents the latest rainfall and river-level data derived from the preprocessed wet-day dataset, while Figure 2 shows the distribution of historical flooding across the UK filtered by postcode. Figure 3 visualises UK flood risk within local authority boundaries and overlays the most recent river-level measurements, and Figure 4 similarly maps flood risk by local authority but is layered with the latest rainfall information taken from both typical-day and wet-day datasets.

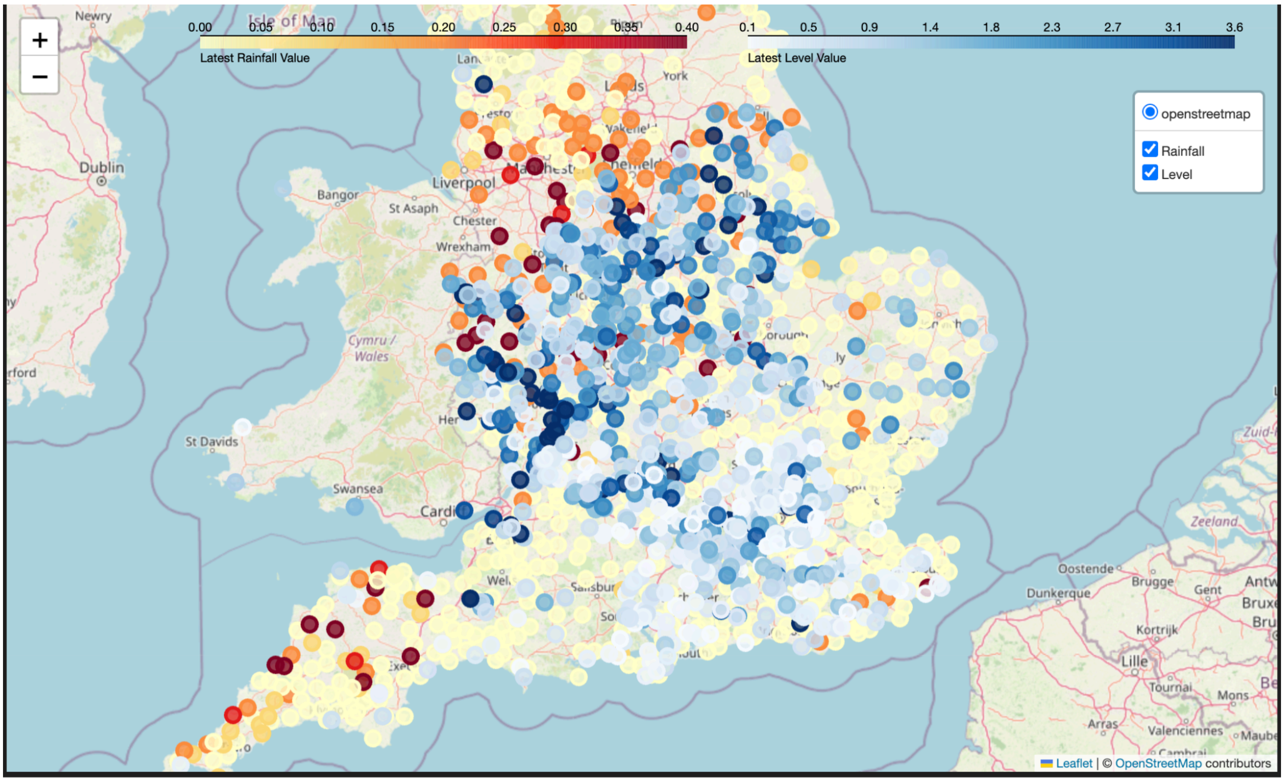

Figure 1. Latest Rainfall and River Level data from preprocessed wet day data.

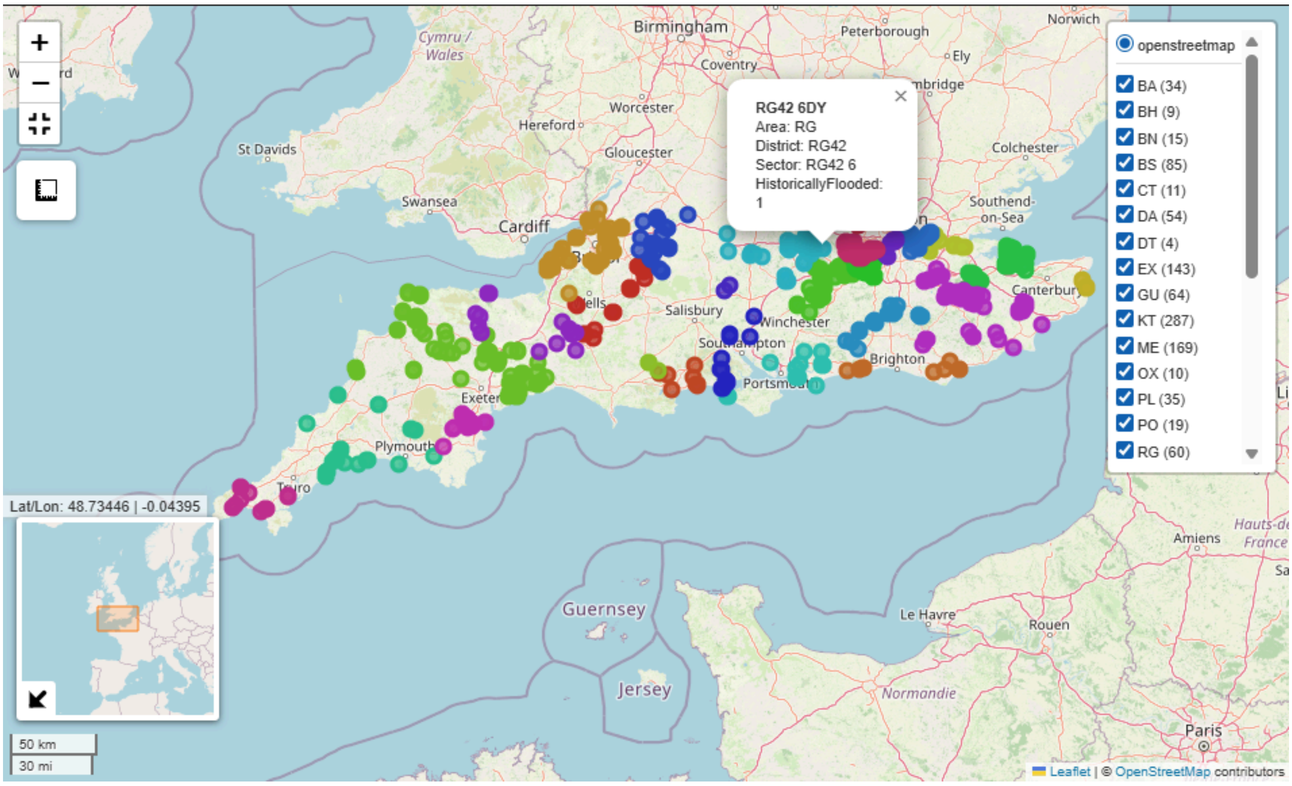

Figure 2. Map of Historical Flooding in the UK, filtered by postcode

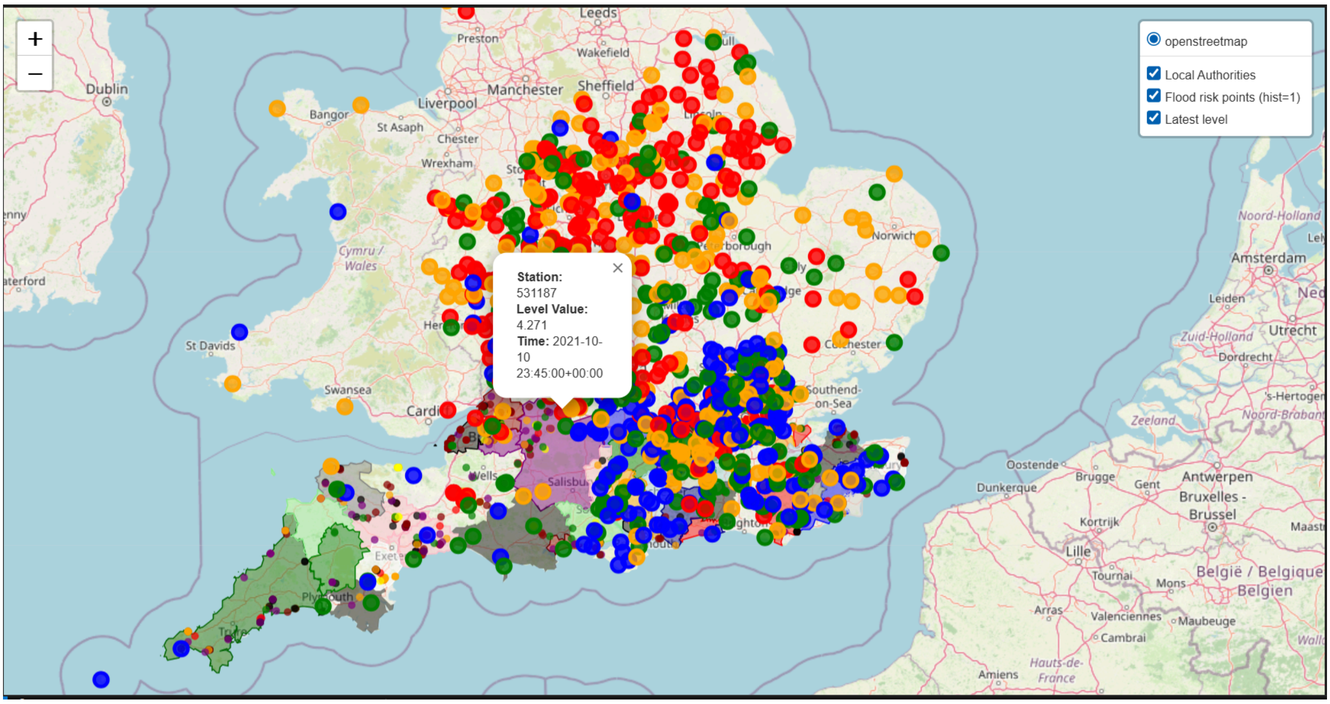

Figure 3. Local Authority and Flood Risk with the Latest River Level Data (Wet Day + Typical Day)

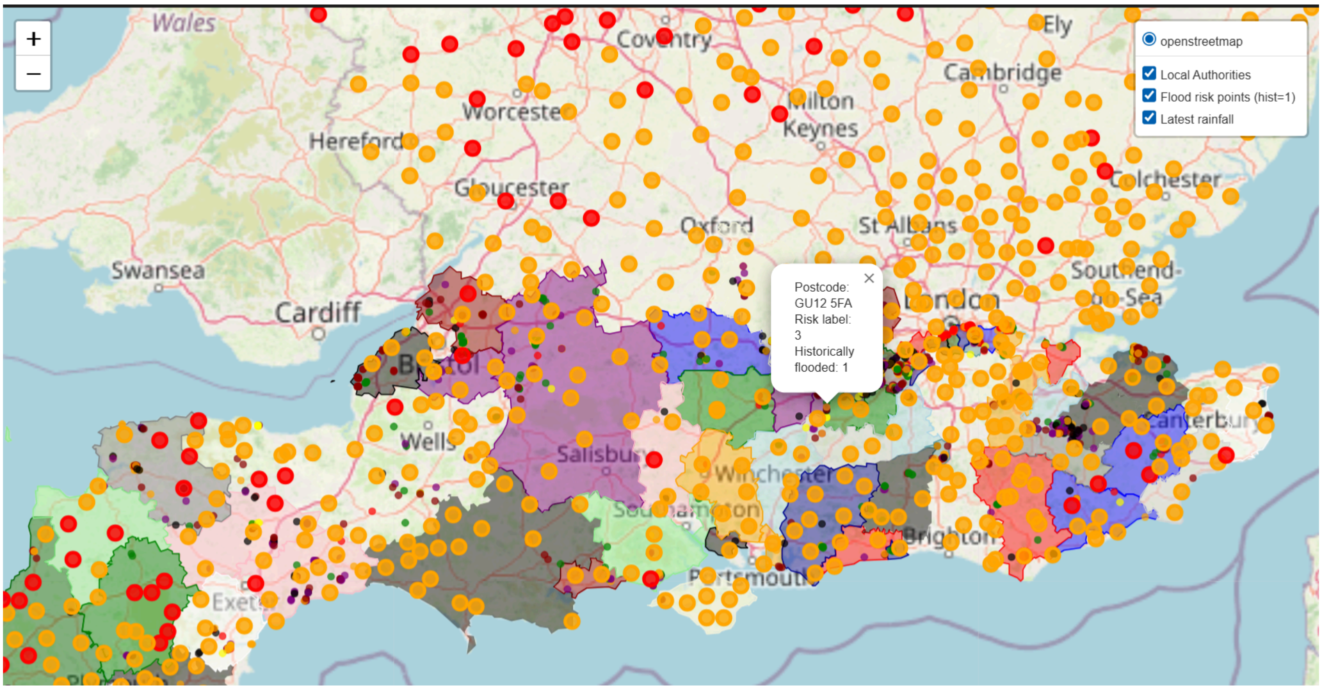

Figure 4. Local Authority and Flood Risk with the Latest Rainfall Data from Wet and Typical Days